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Subject: Web Application and Development	Sem: III AY: 2026-27
EXPERIMENT No: 12	Assignment No: 12
TITLE: To understand the basic use of Git, Jenkins, and Apache Tomcat for managing and running a web application.	

Task

Create a simple web application and perform the following:

1. Create a **GitHub repository** and upload your application code using Git commands.
2. Clone the repository and make one small change to the application.
3. Use **Jenkins** to fetch/check the code from the GitHub repository.
4. Install and start **Apache Tomcat 8 or above**.
5. Deploy and run the web application on Tomcat.
6. Open the application in a web browser and verify that it is working.

Yes. For a **simple student-level assignment**, I recommend using a basic **HTML web application** and deploying it on Tomcat as a **WAR file**. This avoids unnecessary Java/JSP complexity while still demonstrating Git, GitHub, Jenkins, and Tomcat.

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1. Create a folder :

   D:\SIT\WAD\E12\GitJenkinsTomcat\MyWebApp

2. Initialize Git
   git init
3. Now tell Git to track your file:
   git add .

   then
   git commit -m "Initial web application"

```
PS D:\SIT\WAD\E12\GitJenkinsTomcat\MyWebApp> git init
>> git add .
>> git commit -m "Initial web application"
Initialized empty Git repository in D:/SIT/WAD/E12/GitJenkinsTomcat/MyWebApp/.git/
[master (root-commit) c452932] Initial web application
 1 file changed, 1 insertion(+)
 create mode 100644 index.html
PS D:\SIT\WAD\E12\GitJenkinsTomcat\MyWebApp>
```

Now, Create GitHub Repository

Go to: GitHub

Login. Click: **+ → New repository**

Give repository name: MyWebApp

Select: **Public.**

Connect your local project to GitHub

On your computer, on the command prompt:

git remote add origin https://github.com/vatsalposty01/MyWebApp.git

git branch -M main

Then:

git push -u origin main

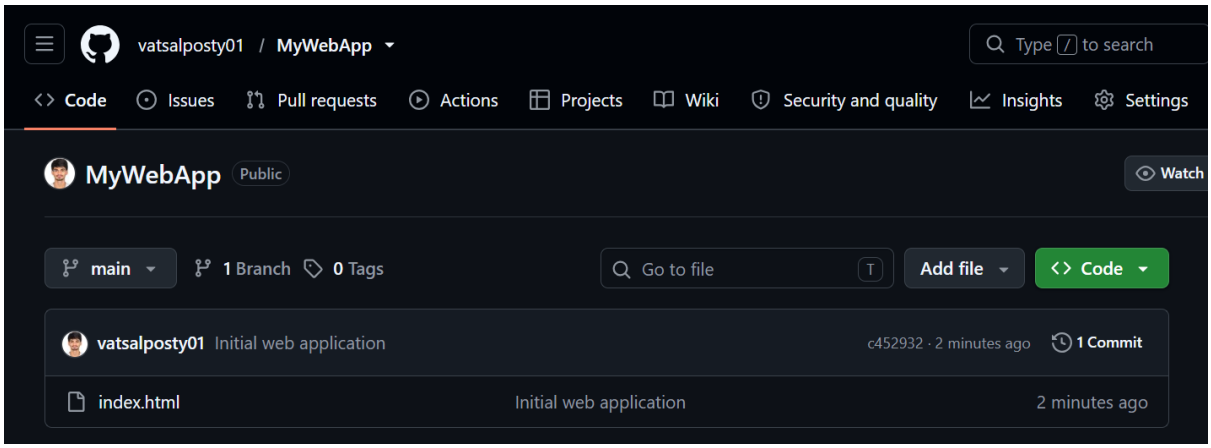
GitHub may ask you to sign in.

After successful upload, refresh your GitHub repository.

You should see:

index.html

```
PS D:\SIT\WAD\E12\GitJenkinsTomcat\MyWebApp> git remote add origin https://github.com/vatsalposty01/MyWebApp.git
>> git branch -M main
>> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 240 bytes | 240.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/vatsalposty01/MyWebApp.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS D:\SIT\WAD\E12\GitJenkinsTomcat\MyWebApp>
```



Now, lets Clone the repository

Go to your Desktop.

Open Command Prompt.

Now clone your GitHub repository:

```
git clone https://github.com/vatsalposty01/MyWebApp.git
```

You will get a new folder:

```
Desktop
|
└── MyWebApp
    |
    └── index.html
```

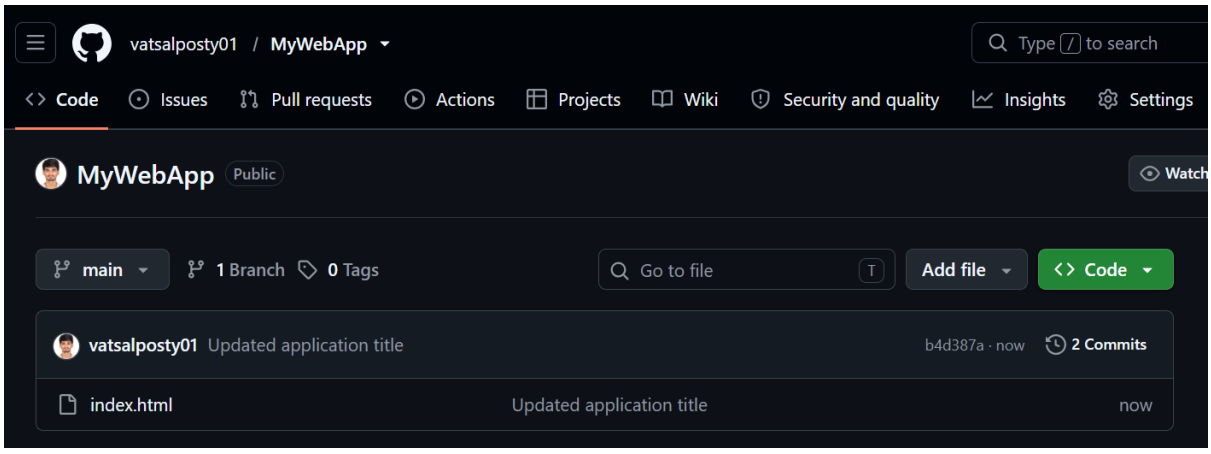
Open index.html **in the original folder** and make a small change into it.

Open Command Prompt in the MyWebApp folder.

```
git add .
git commit -m "Updated application title"
git push
```

Now refresh GitHub.

The updated code will be visible there.



Install Jenkins

Download Jenkins from the official Jenkins website:

Jenkins

Install Jenkins using the normal/default options.

During installation, Jenkins will ask for Java/JDK. **(install jdk 25 from https://www.oracle.com/in/java/technologies/downloads/#jdk25-windows)**

Make sure Java is installed.

After installation, Jenkins normally runs at:

http://localhost:8080

Open Jenkins

You will see the Jenkins login/unlock screen.

Jenkins may ask for an **Administrator Password** during first setup.

The password is normally stored in a file similar to:

C:\ProgramData\Jenkins\.jenkins\secrets\initialAdminPassword

Open that file and copy the password.

Paste it into Jenkins.

Select:

Install suggested plugins

Wait for installation.

Create your Jenkins user.

Create a Jenkins Job

In Jenkins:

Click:

New Item

Enter:

MyWebApp

Select:

Freestyle project

Click:

OK

Tell Jenkins where the GitHub code is

In the Jenkins job:

Find:

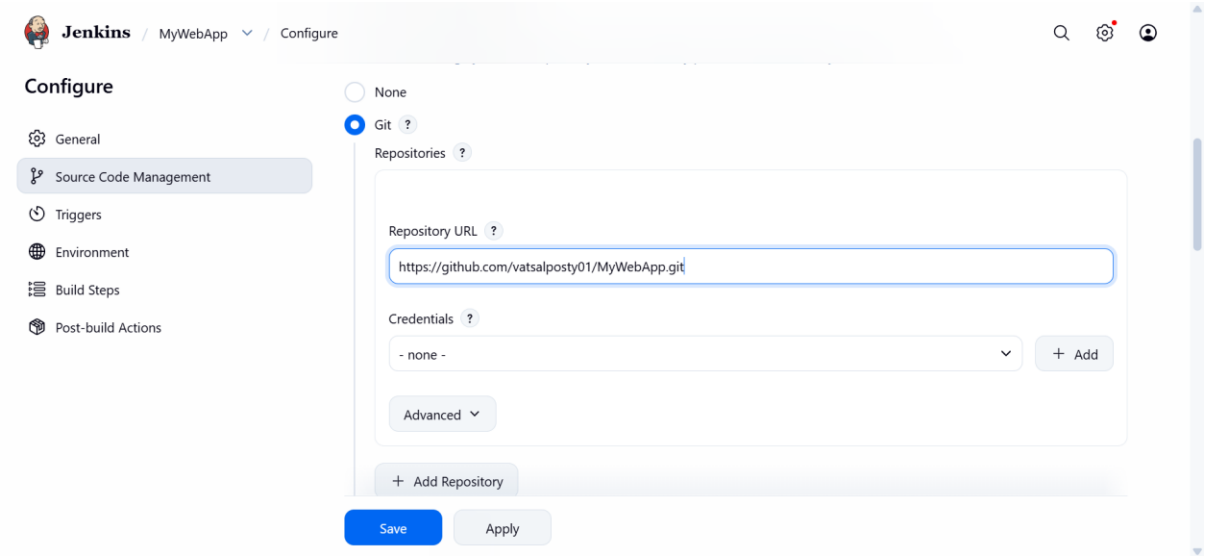
Source Code Management

Select:

Git

In Repository URL enter add: https://github.com/vatsalposty01/MyWebApp.git

If your repository is public, you generally don't need credentials.



Click:

Save

Run Jenkins

Open your Jenkins project:

MyWebApp

Click:

Build Now

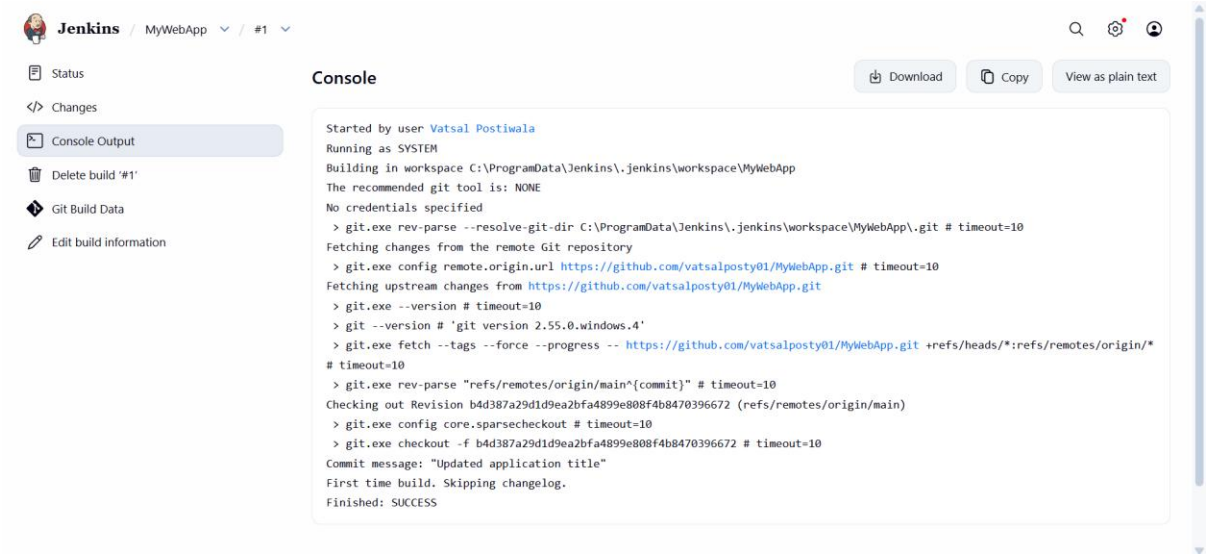
Jenkins will download/check the code from GitHub.

On the left side you will see something like:

Build #1

Click:

Build #1 → Console Output



You should see messages indicating that Jenkins cloned/fetched the Git repository.

This completes:

Use Jenkins to fetch/check the code from the GitHub repository.

Install Apache Tomcat

Download Tomcat from the official Apache Tomcat website:

Apache Tomcat

For this assignment, I recommend:

Tomcat 9

Download the **Windows zip** version.

For example:

apache-tomcat-9.x.x.zip

Extract it.

You can put it here:

C:\Tomcat

So you have:

C:\Tomcat

|

├── bin

├── conf

├── lib

├── logs

├── webapps

└── ...

Start Tomcat

Go to Tomcat folder

Open conf

Find server.xml

Open it with notepad

Search for Connector port=”8080”

Change 8080 to 8081

Go to

C:\Tomcat\bin

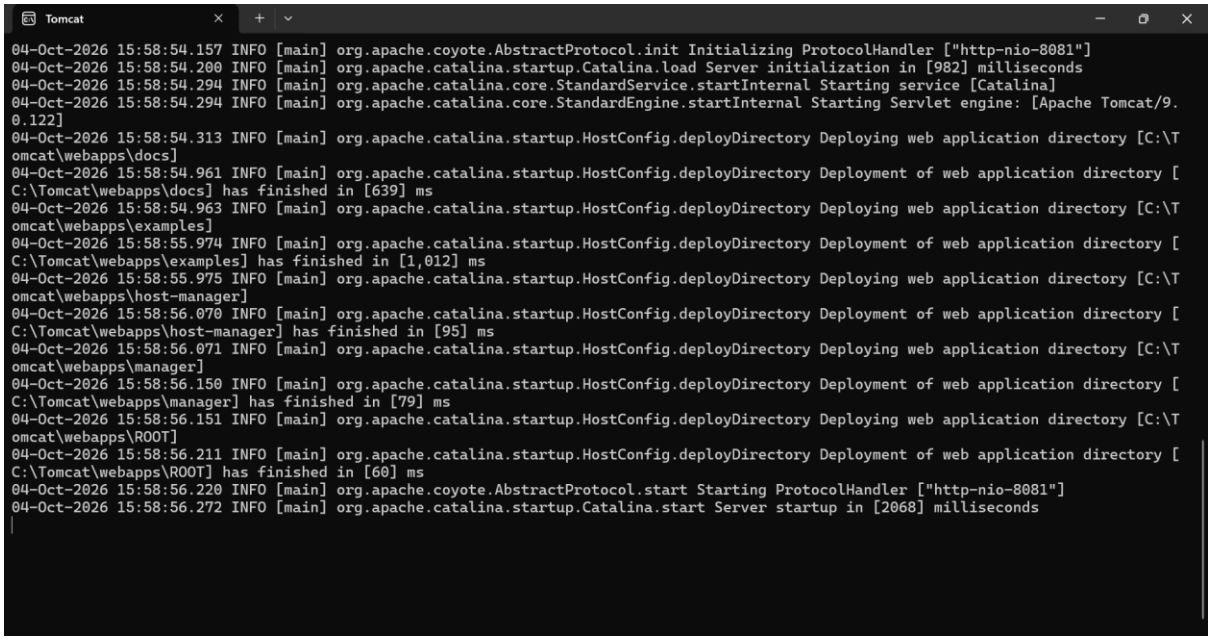
You will see:

startup.bat

Double-click:

startup.bat

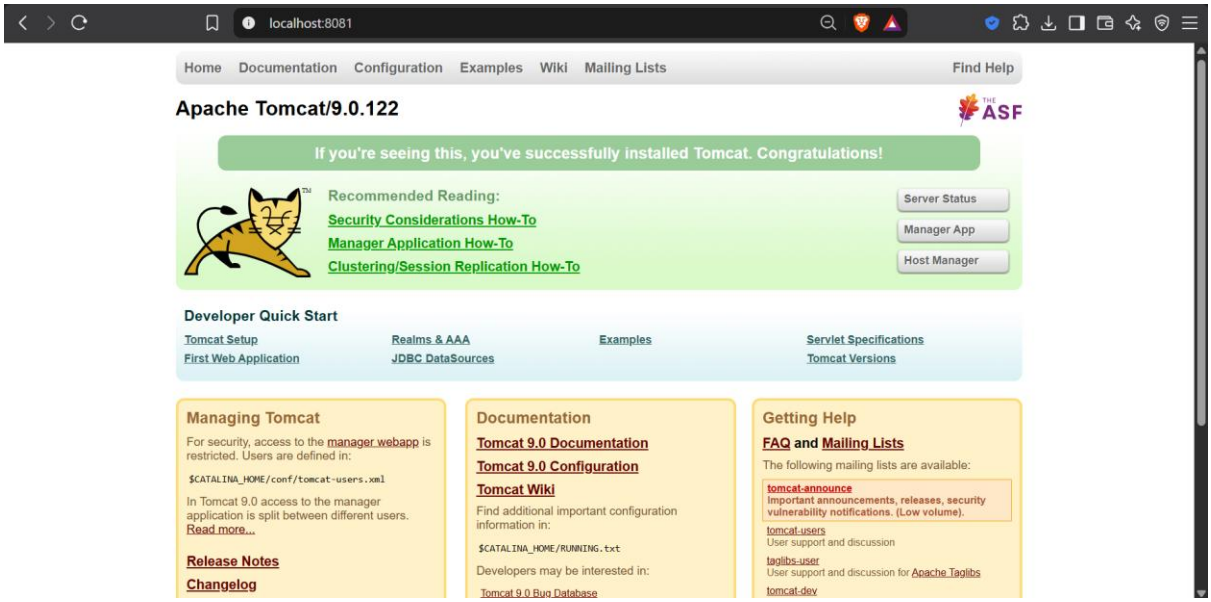
A command window will appear.



Now open Chrome:

http://localhost:8081

If Tomcat is running, you should see the **Tomcat welcome page**.



In case issues are there then check go to command prompt and

java -–version

you should get java version with this.

Again go to command prompt and write

Get-Command java | Select-Object Source

And then

java -XshowSettings:properties -version 2>&1 | Select-String “java.home”

you will get path of jdk

then copy that path and write

$env:JAVA_HOME= the copied path of jdk folder

Again run the startup.bat file in the bin folder of Tomcat by

.\startup.bat

This will start the Tomcat server

Deploy your application

This is the easiest part.

Go to:

C:\Tomcat\webapps

Create a folder:

MyWebApp

Put your:

index.html

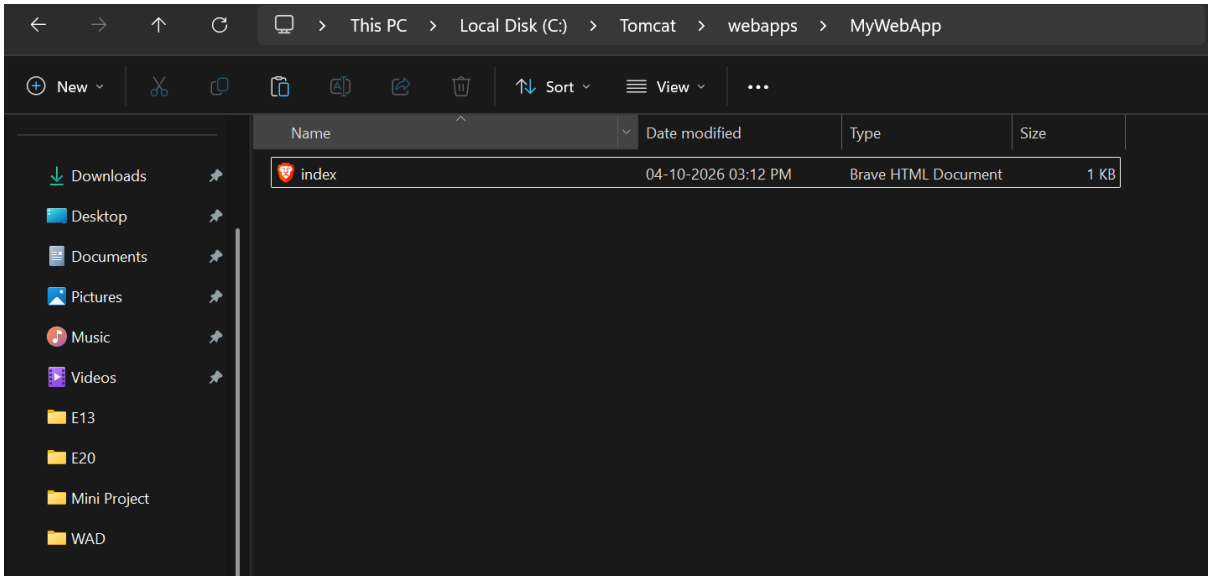
inside it.

So the structure becomes:

C:\Tomcat\webapps\MyWebApp

|

└── index.html



Now execute the application with:

http://localhost:8081/MyWebApp/

